

Midterm Exam: CS 215

Attempt all questions. You have 120 minutes for this exam. Clearly mark out rough work. No calculators or phones are allowed (or required :-)). You may directly use results/theorems we have stated or derived in class, unless the question explicitly mentions otherwise. Avoid writing lengthy answers.

Useful Information

1. The empirical mean of n independent and identically distributed random variables is approximately Gaussian distributed. The approximation accuracy is better when n is larger. If the random variables are Gaussian, the empirical mean is exactly Gaussian distributed.
2. For a non-negative random variable X , we have $P(X \geq a) \leq E(X)/a$ where $a > 0$. This is Markov's inequality.
3. For a random variable X with mean μ and variance σ^2 , we have $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$. This is Chebyshev's inequality.
4. Gaussian PDF: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$, MGF $\phi_X(t) = e^{\mu t + \sigma^2 t^2/2}$
5. Poisson PMF: $P(X = i) = \frac{e^{-\lambda}\lambda^i}{i!}$, MGF $\phi_X(t) = e^{\lambda e^t - \lambda}$
6. Integration by parts: $\int u dv = uv - \int v du$

1. Consider a permutation of the first n positive integers, generated uniformly randomly (i.e. each of the $n!$ different permutations are equally likely). The ordered pair (i, j) in the permutation is called an inversion if $i < j$ but j precedes i (i.e. occurs earlier than i) in the permutation. Determine the expected number of inversions in a uniformly randomly generated permutation of the first n positive integers. [10 points]

2. This problem concerns the design of a spam filter based on knowledge of basic discrete probability. You have a 'training set' of 2000 spam messages and 1000 non-spam messages. A word 'ABC' appears in 400 spam and 60 non-spam messages in the training set. Likewise, the word 'PQR' appears in 200 spam and 25 non-spam messages. Multiple occurrences of a word in the same message are counted as just one. Let E_1 and E_2 denote the events that a message contains the words 'ABC' and 'PQR' respectively. Let S be the event that a message is spam. Assume (i) that E_1 and E_2 are independent, (ii) that $E_1|S$ and $E_2|S$ are also independent, and (iii) that $P(S) = P(S^c)$ where S^c is the set-complement. Estimate the probability that a new message (not in the training set) that contains both the words 'ABC' and 'PQR' is a spam message. (You can use the training set to estimate certain probability values). [10 points]

3. Consider independently drawn sample values $x_1, x_2, \dots, x_n$, each from $\text{Poisson}(\lambda/n)$ where n is known. What is the maximum likelihood estimate for λ ? Derive the bias, variance, MSE of this estimator. Is this a consistent estimator? Why (not)? [10 points]

(There is a physical significance to this question, even though one needn't understand it to answer the question. The noise in an image pixel is typically Poisson in nature. The values $x_1, x_2, \dots, x_n$ correspond to n images of the same scene acquired in quick succession with acquisition time T/n per image, instead of acquiring one image in time T .)

4. If $X \sim \mathcal{N}(0, 1)$, then prove that $P(|X| \geq u) \leq \sqrt{2/\pi} \frac{e^{-u^2/2}}{u}$ for all $u > 0$. How does this bound compare with that given by Chebyshev's inequality? [10+5 = 15 points]
5. Consider n values $\{x_i\}_{i=1}^n$ drawn independently from a Laplacian distribution with mean 0 and parameter σ . The probability density for a Laplacian random variable X is given by $f_X(x) = \frac{1}{2\sigma} e^{-|x|/\sigma}$ (note the absolute value in the exponent). Given $\{x_i\}_{i=1}^n$, derive the maximum likelihood estimate for σ , as well as its bias, variance, MSE. [15 points]
6. In this problem, we will derive higher order moments of specific random variables in a new way.
- (a) Consider $X \sim \mathcal{N}(\mu, \sigma^2)$. Then prove that $E[g(X)(X - \mu)] = \sigma^2 E[g'(X)]$ where g is a differentiable function such that $E[|g'(X)|] < \infty, |g(x)| < \infty$. Use this to derive an expression for $E[X^3]$ in terms of μ and σ^2 . Do not use any other method (eg: MGFs) to derive $E[X^3]$. [5+5=10 points]
- (b) Consider $X \sim \text{Poisson}(\lambda)$. Then prove that $E[\lambda g(X)] = E[Xg(X-1)]$ where g is a function such that $-\infty < E[g(X)] < \infty, -\infty < g(-1) < \infty$. Use this to derive an expression for $E[X^3]$ assuming known expressions for $E[X], E[X^2]$. Do not use any other method (eg: MGFs) to derive $E[X^3]$. [5+5=10 points]
7. (a) A student is trying to design a procedure to generate a sample from a distribution function F , where F is invertible. For this, (s)he generates a sample u_i from a $[0, 1]$ uniform distribution using the 'rand' function of MATLAB, and computes $v_i = F^{-1}(u_i)$. This is repeated n times for $i = 1 \dots n$. Prove that the values $\{v_i\}_{i=1}^n$ follow the distribution F . [6 points]
- (b) Let $Y_1, Y_2, \dots, Y_n$ represent data from a continuous distribution F . The empirical distribution function F_e of these data is defined as $F_e(x) = \frac{\sum_{i=1}^n \mathbf{1}(Y_i \leq x)}{n}$ where $\mathbf{1}(z) = 1$ if the predicate z is true and 0 otherwise. Now define $D = \max_x |F_e(x) - F(x)|$. Also define $E = \max_{0 \leq y \leq 1} \left| \frac{\sum_{i=1}^n \mathbf{1}(U_i \leq y)}{n} - y \right|$ where $U_1, U_2, \dots, U_n$ represent data from a $[0, 1]$ uniform distribution. Now prove that $P(E \geq d) = P(D \geq d)$. Briefly explain what you think is the practical significance of this result in statistics. [8+6=14 points]